CV

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Namhyun Kim

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Website Google Scholar

Research Interests

Beyond-5G/6G communication systems; massive MIMO; integrated sensing and communication (ISAC); AI/ML-based communications; wireless foundation models; semantic communications; Bayesian estimation.

Education

Arizona State University (ASU), Tempe, AZ, USA — Jan. 2025 – Present
Doctor of Philosophy in **Electrical, Computer and Energy Engineering**
Advisor: Prof. Ahmed Alkhateeb.

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea — Mar. 2021 – Feb. 2023
Master of Science in **Electrical Engineering** — Cumulative GPA: 3.81/4.30
Dissertation: Location-Aware Downlink Beamforming with Marginal Training Overhead for MmWave FDD Massive MIMO Systems.
Advisor: Prof. Junil Choi.

Yonsei University, Seoul, South Korea — Mar. 2015 – Feb. 2021
Bachelor of Science in **Electrical Engineering** (High Honors) — Cumulative GPA: 4.04/4.30

Professional Experience

InterDigital, Inc. — Philadelphia, PA, USA
Ph.D. Intern, Wireless Lab — June 2026 – Aug. 2026

- Calibrated integrated sensing and communication (ISAC) channel models based on 3GPP TR 38.901 for joint sensing and communication simulations.
- Developed a semantic communication framework incorporating 3GPP TR 38.901 spatial consistency procedures to capture channel evolution across user locations.

SK Telecom, Co., Ltd. — Seoul, South Korea
Manager, LTE/5G Radio Access Network (RAN) Performance Improvement Group — Jan. 2022 – Sep. 2023

- Performed LTE/5G L1/L2/L3 network planning, engineering, troubleshooting, and performance analysis.
- Conducted commercial verification testing of 64-TRX massive MIMO systems with Samsung Electronics, Co., Ltd.
- Used Python, SQL, Kubernetes, Linux, and MATLAB for network engineering and analysis.

Publications

Manuscripts under Revision and Preprints

- **N. Kim**, S. Alikhani, and A. Alkhateeb, "LWM-TF: A Wireless Foundation Model on 5G NR Time-Frequency Resource Grids," *IEEE Journal of Selected Topics in Signal Processing (JSTSP)*, **under revision**. [arXiv].
- **N. Kim**, M. S. Abouamer, J. Park, P. Popovski, and A. Alkhateeb, "Digital Twin-Aided Prescreening for User Scheduling in MU-MIMO Downlink Systems," arXiv preprint arXiv:2608.10364, 2026. [arXiv].

Journal Articles

- **N. Kim**, J. Han, J. Choi, A. Alkhateeb, C.-B. Chae, and J. Park, "Integrated Sensing and Communications in Downlink FDD MIMO Without CSI Feedback," *IEEE Transactions on Wireless Communications*, 25:2984–3000, 2026. [Link].
- J. Han, **N. Kim**, and J. Park, "Reducing Latency by Eliminating CSIT Feedback: FDD Downlink MIMO Transmission for Internet-of-Things Communications," *IEEE Internet of Things Journal*, 13(6):11301–11315, 2026. [Link].
- **N. Kim**, I. P. Roberts, and J. Park, "Splitting Messages in the Dark—Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback," *IEEE Transactions on Wireless Communications*, 24(4):3320–3332, 2025. [Link].

Conference Papers

- **N. Kim**, S. Alikhani, and A. Alkhateeb, "LWM-Spectro: A Foundation Model for Wireless Baseband Signal Spectrograms," *IEEE International Conference on Communications (ICC)*, Glasgow, UK, pp. 1–6, 2026. [Link].
- **N. Kim** and J. Park, "Robust Precoding and Rate-Splitting Multiple Access for FDD Massive MIMO Without CSI Feedback," *IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, pp. 1–6, 2025. [Link].
- **N. Kim**, J. Han, and J. Park, "Integrated Sensing and Communications in FDD MIMO Without CSI Feedback: Towards FDD MIMO ISAC," *22nd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, Seoul, South Korea, pp. 132–137, 2024. [Link].

Projects

LTE Link-Level Simulation Seoul, South Korea
Intelligent & Secure underwater CommunicaTion (InSeCT), KRIT Sep. 2023 – Apr. 2024

- Implemented the signal processing chain in an OFDM receiver.
- Derived a post-equalization SINR model for CFO-impaired OFDM transmission.
- Linked post-equalization SINR to BICM capacity.

5G Shared Network Commercial Deployment Daejeon, South Korea
SK Telecom Sep. 2022 – Aug. 2023

- Expanded 5G coverage in Korea through shared-network deployment with KT and LGU.
- Integrated and optimized network equipment while addressing security concerns related to shared infrastructure.
- Managed traffic loads and coordinated RAN operations to maintain service quality.

Undergraduate Capstone Project Seoul, South Korea
School of Electrical Engineering, Yonsei University Mar. 2020 – Aug. 2020

- Completed a department-led capstone project during the COVID-19 pandemic.
- Developed a campus access control system combining RFID and temperature measurement sensors.

Scholarships & Awards

Global Korea Scholarship: Study Abroad Scholarship Program NIIED, Korean Government
National scholarship supporting graduate studies abroad. 2025–2026

Graduate Student Academic Excellence Scholarship Wooyang Foundation
Merit-based scholarship recognizing academic excellence. 2021–2022

Korean Government-funded Student KAIST
Master's studies fully funded by Korean government scholarships. 2021–2022

Department Chair Commendation Dept. of EE, Yonsei University
Recognized for tutoring international students at Songdo International Campus. Feb. 2021

Academic Excellence Award Yonsei University
Semester Honors (Spring 2016, Fall 2018, Spring 2019), High Honors (Spring 2020), Highest Honors (Fall 2019).

Graduation with Distinction Yonsei University
Graduated with High Honors, placing in the top 3% of electrical engineering graduates. Feb. 2021

Teaching and Mentoring Experience

Graduate Teaching Assistant Daejeon, South Korea
Course: EE528: Engineering Random Processes, KAIST Sep. 2021 – Feb. 2022

- Served as a teaching assistant for Prof. Junil Choi's Engineering Random Processes course.

Academic Tutor for International Students Songdo, Incheon, South Korea
EE Honor Society Program, Dept. of Electrical Engineering, Yonsei University Sep. 2019 – Feb. 2021

- Tutored 21 international students in Engineering Mathematics and Engineering Physics.
- Conducted tutoring sessions at students' request; most participants achieved strong grades.

Skills

- **Programming:** Python, C/C++, MATLAB, SQL.
- **Software:** Linux, Kubernetes.
- **Soft Skills:** Communication, adaptability, problem-solving.